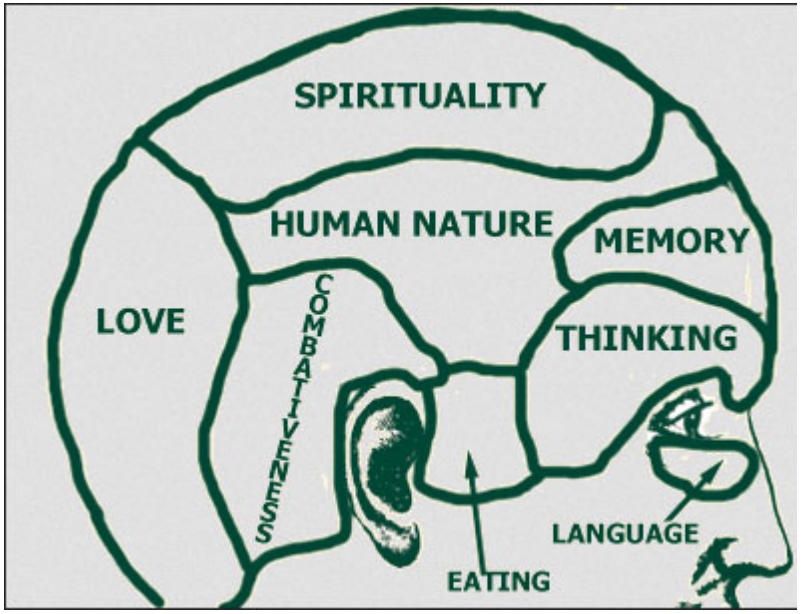
Brains and Language: Broca to today.

LING/PSYC 406 Session 5
(Covering Finger's biographical sketch of Broca, and Posner et al. (1988))

An early attempt at relating structure to function: Phrenology



A simplified phrenological chart.

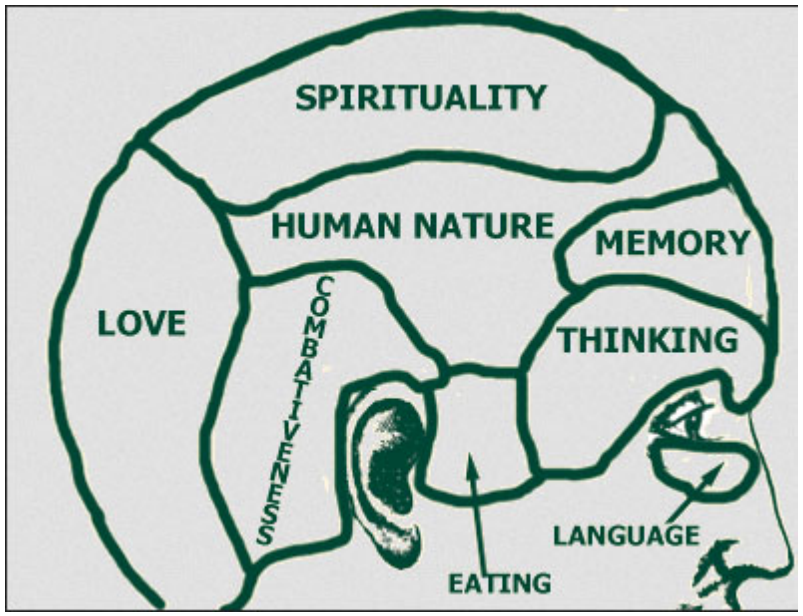
The premise:
1. different parts of the brain are responsible for different processes.
2. aptitude is related to differences in brain region size.
3. brain region size influences skull shape.

Therefore:

Careful measurements of people's skulls will reveal relationships between skull shape and various brain functions.

No one believes in phrenology anymore.
But which of these premises are actually *completely and obviously wrong*?

An early attempt at relating structure to function: Phrenology

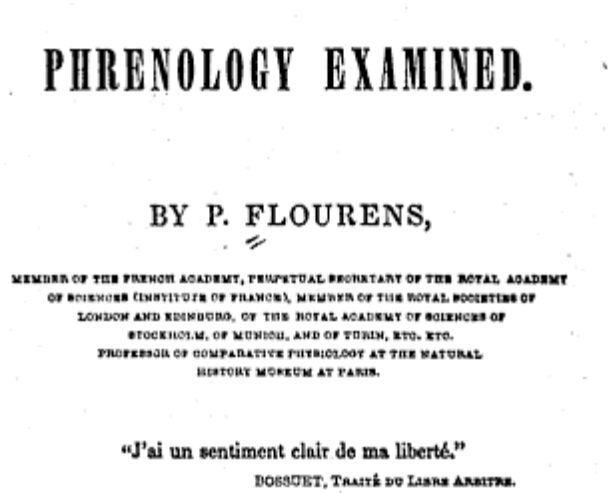


A simplified phrenological chart.

Important problems for phrenology:
1. inappropriate *anatomical* measurement –
skull shape not a sensitive measure of brain shape.

2. unconstrained definition of
functions: e.g., studies of prisoners aimed identifying regions associated with “criminality,” or women with different numbers of children to find the locus of “philopropogenitiveness.”

Experimental lesions and a critique of Phrenology



PHRENOLOGY EXAMINED.

BY P. FLOURENS,

MEMBER OF THE FRENCH ACADEMY, PERPETUAL SECRETARY OF THE ROYAL ACADEMY OF SCIENCES (INSTITUTE OF FRANCE), MEMBER OF THE ROYAL SOCIETIES OF LONDON AND EDINBURGH, OF THE ROYAL ACADEMY OF SCIENCES OF STOCKHOLM, OF MUNICH, AND OF TURIN, ETC. ETC.
PROFESSOR OF COMPARATIVE PHYSIOLOGY AT THE NATURAL HISTORY MUSEUM AT PARIS.

"J'ai un sentiment clair de ma liberté."
BOSSUET, TRAITÉ DU LIBRE ARBITRE.

Flourens's anti-phrenology pamphlet.
(available here)

Flourens was an early and vociferous critic of Phrenology.

His experiments (mainly with pigeons and rabbits) established distinct functions of cerebral cortex, cerebellum, and mid-brain structures.

BUT, he failed to find any correspondence between specific cortical regions and specific senses.

Instead, he insisted that the *size* of the lesion was critical, and that when one sense was lost, all were lost together (e.g., as when most of cortex is removed and sight, hearing, etc. disappear together).

Localization and language (aphasia)



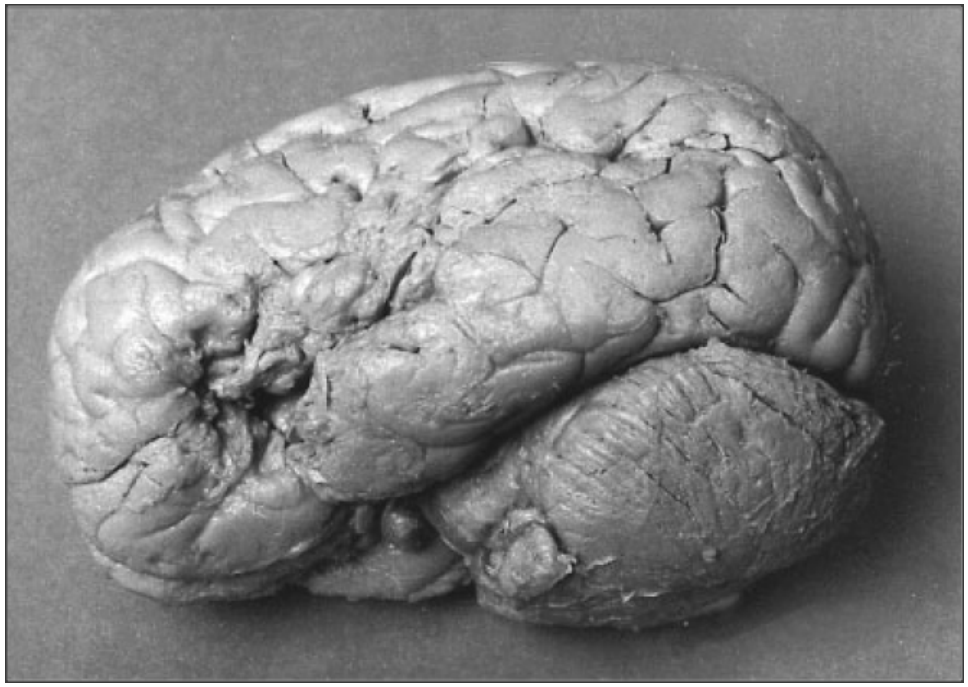
In 1861, neurologist Pierre Paul Broca presented the case of "Tan" who had lost the ability to speak after a brain injury decades earlier.

He gave the presentation to the newly established Society of Anthropology, of which he was also the founding president.

Tan:

- could only produce one jargon syllable
- seemed otherwise to be "in possession of his senses"
- *in fact, seemed to understand what was being said.*

This was not the first observation of a specific loss of





Tan's brain on display at the Musee Dupuytren

Anatomical description:

- large lesion

- definitely including the third convolution of the frontal lobe (inferior frontal gyrus)

- more details on this in a moment...

Thus, we have "Broca's area:"
a portion of the left, frontal lobe that is important for speech.

Consider this contemporary criticism:

- "to locate the damage which destroys speech and to locate speech are two different things" Hughlings Jackson (1874)

Localization and language (aphasia)

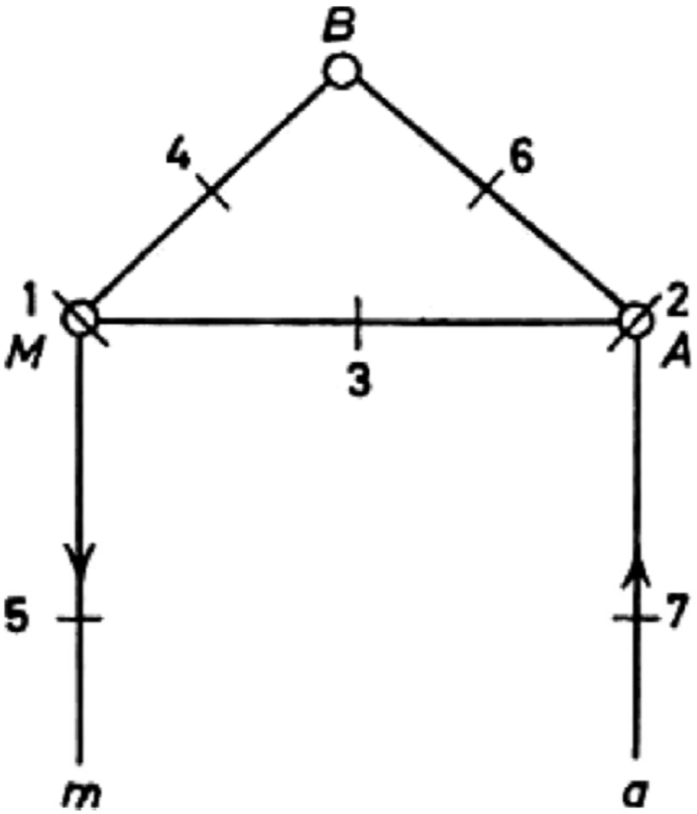


Fig. 36.5. Lichtheim's "House" model. A represents the center for sound images; M is the center for movement images; B is the center for concepts; a is the acoustic pathway and m is the motor pathway. 1–7 represent various forms of aphasia, depending on the destruction of a center or pathway.

In the 1870's a number of German neurologists — Lichtheim, Wernicke — began making more systematic observations of aphasia.

They ascribed specific losses of function to "disconnections" between "centers."

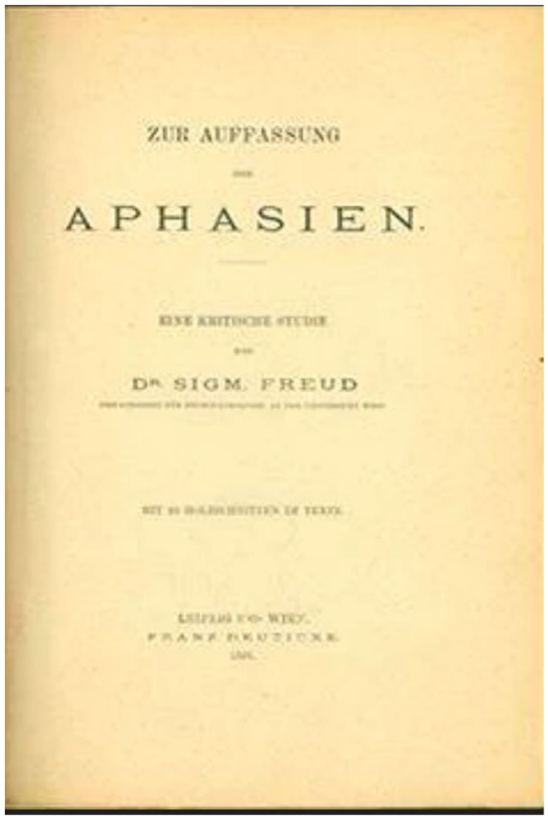
So, a loss of the ability to understand spoken words might be due to a disconnection between "A" and "B."

A loss of the ability to do verbatim repetition may be due to a disconnection between "A" and "M," and so on.

This approach of describing "centers" for specific functions is an ancestor of a topic we'll be dealing with later on in the course: "modularity"

Eling, P., & Whitaker, H. (2009). History of aphasia: from brain to language. In *Handbook of clinical neurology* (Vol. 95, pp. 571-582).

A holist critique


ZUR AUFFASSUNG

APHASIEN.

EINE KRITISCHE STUDIE

Dr SIGM. FREUD

Sigmund Freud (!) critiqued the “diagram-drawing” school of aphasia research in his early neurological research.

He felt it was naïve to expect the brain to be organized according to psychological categories like “word forms” or “concepts.”

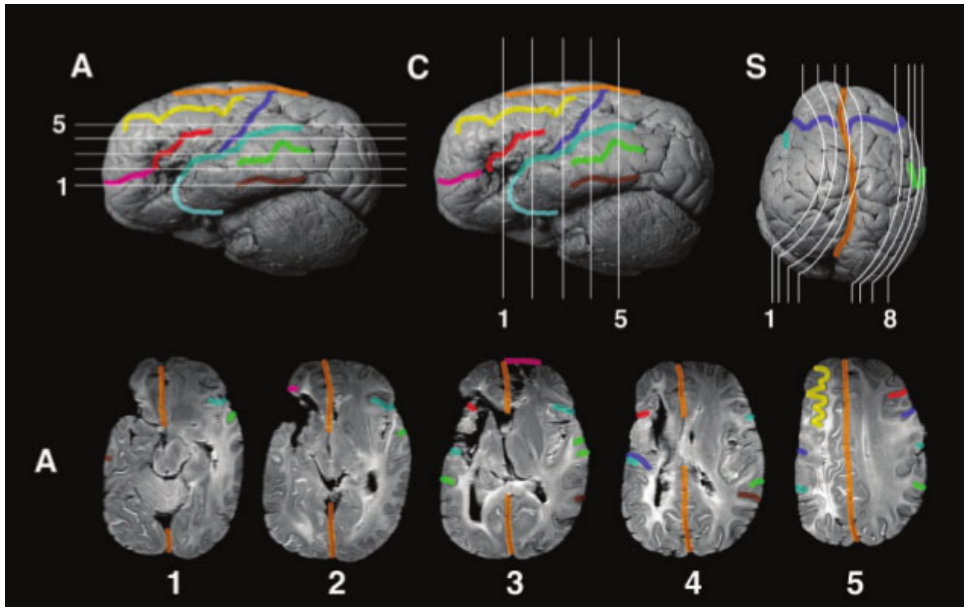
He proposed instead that language emerged from “fields” connecting perceptual and motor abilities to “drives.”

This is a durable tension in cognitive neuroscience:

*Can we directly relate **brain area** functions and **cognitive** functions or not?*

Is the brain really organized according to our descriptions of cognitive phenomena?

Back to Leborgne's brain...



Using MRI, Dronkers and colleagues were able to assess the extent and location of damage in Broca's landmark cases very precisely.

The lesion is very deep, and includes the insula.

There is also a lot of white matter damage to the superior longitudinal fasciculus (which, basically connects the front and back of the cerebral cortex).

Indeed, other work by this group suggests that the insula, not Broca's area, is strongly associated with *speech apraxia* (the condition Broca described in his patients).

Occasional Paper:
N. F. Dronkers, O. Plaisant, M. T. Iba-Zizen, and E. A. Cabanis
Paul Broca's historic cases: high resolution MR imaging of the brains of Leborgne and Lelong
Brain (2007) 130 (5): 1432-1441 first published online April 2, 2007 doi:10.1093/brain/awm042

Why are we still talking about Broca 150 years later?

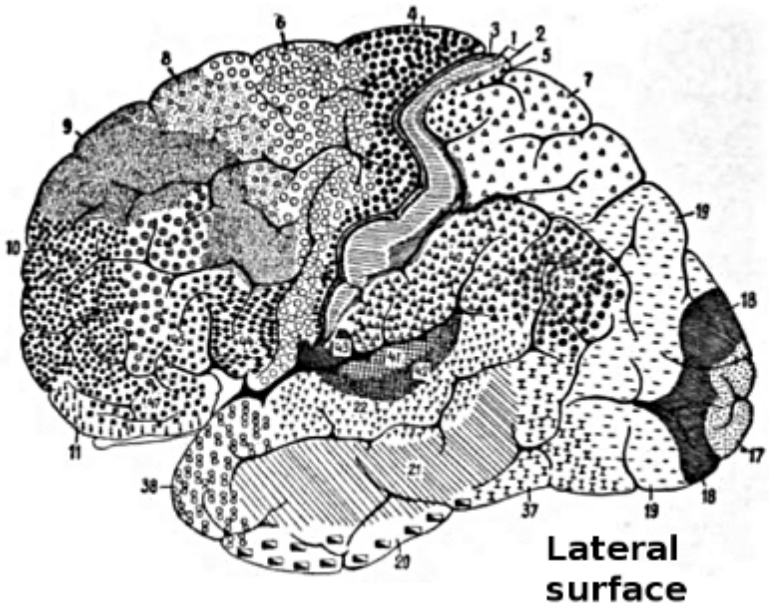
His characterization of the behavior was crude.

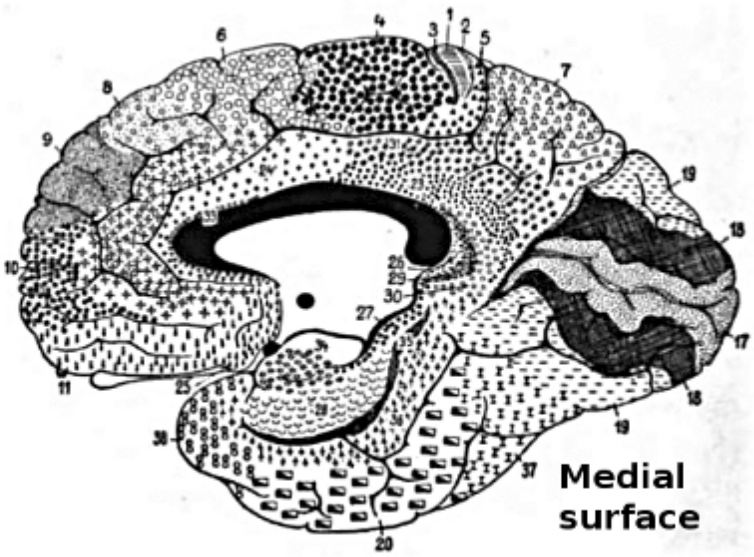
- But at the time, linguistics was very much focused on historical issues, understanding how languages were related to each other, and reconstructing ancient languages.

- Further, experimental psychology hadn't really been invented yet. Observations of behavior were necessarily informal.

His anatomical localization was also crude. And probably wrong.

- Relatively fine-grained anatomical descriptions of the cerebral cortex were still in their infancy. For example, Brodmann's map wasn't published until 1909.





Why are we still talking about Broca 150 years later?

Finally...there were other people such as the Daxes, and Wernicke, showing similar things at around the same time.

(If anything Wernicke's data are more interesting because his patient was fluent, but produced meaningless gibberish.)

But for a variety of reasons, Broca's presentation of Tan is seen as a watershed moment in *neuropsychology generally.*

It turned the project of relating specific functions to specific brain regions into a respectable pursuit.



Or, at least, to the more recent past...

Cognitive Neuropsychology depends on Cognitive Psychology for definitions of *functions*.

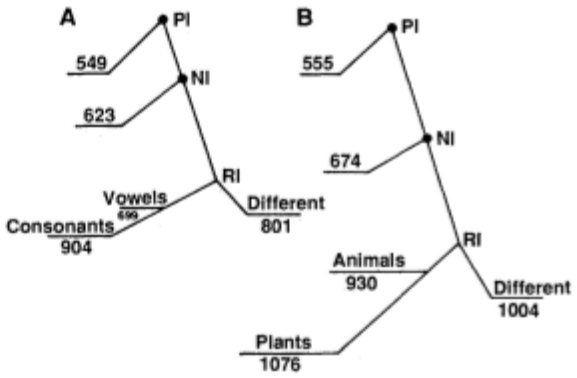


Fig. 1. Results of reaction time studies in which subjects were asked to classify whether pairs of letters were either both vowels or both consonants (**A**) or whether pairs of words were both animals or both plants (**B**). Reaction times are in milliseconds. Each study involved 10 to 12 normal subjects. Standard deviations are typically 20% of the mean value. Data argue in favor of these matches being made on different internal codes (*6, 7*). Abbreviations: PI, physical identity; NI, name identity; and RI, rule identity. [Reprinted from (*8*) with permission of MIT Press]

Much of what we learned about structure-function relationships in the 20th Century came from *experimental* observations of individuals with brain damage.

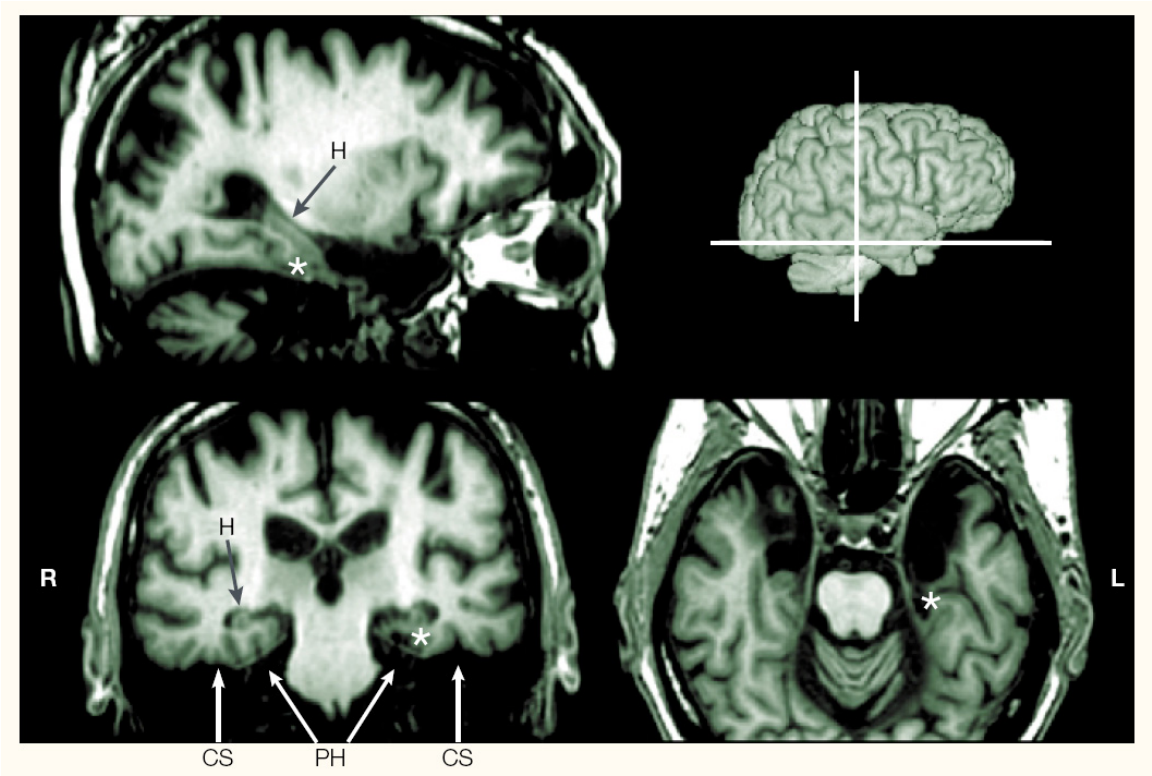
Theory about cognitive processes provides hypotheses about what operations might be separable, e.g.,

Can you dissociate the ability to recognize the abstract forms of words from the ability to compute their meaning?

The methods provide a way to test this:

Quantitative measurements provide a means for statistical comparison. This is critical b/c deficits are rarely **complete.**

Anatomical imaging permits high-resolution location of brain injuries.



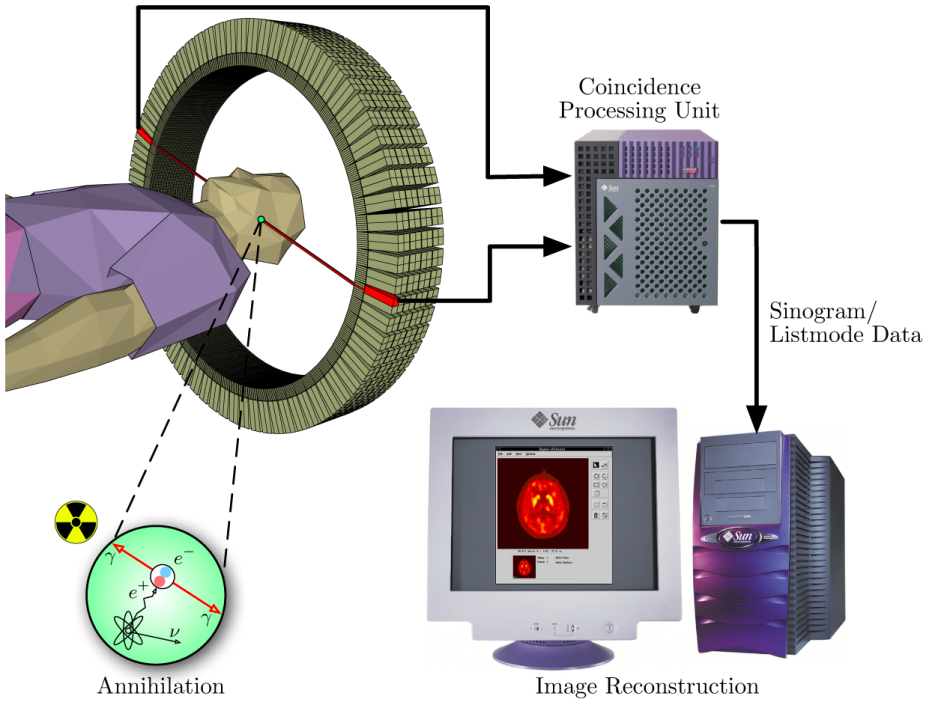
Scan of HM's middle temporal lobes...
(note this was the result of surgery)

Advances in imaging technology meant scientist no longer had to wait for patients to "come to autopsy" before determining the site of their lesion.

But remember: "To locate the damage which destroys speech and to locate speech are two different things" Hughlings Jackson (1874)

What we learn from studies of brain-injured patients is what behavior looks like after a region is damaged. It can only tell us indirectly about what the region might actually *do* when *intact.*

Metabolic neuroimaging allows us to study brains *in action.*



Posner *et al.* describe the results of PET studies, which used radio-labeled oxygen as a marker for activity.

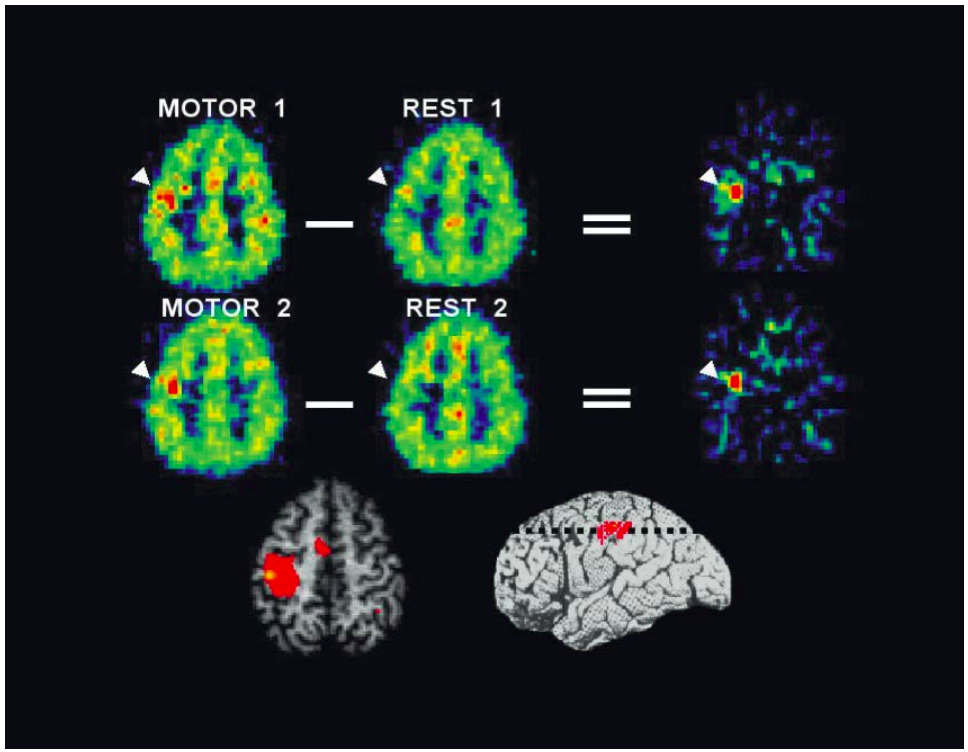
Some important things to understand about how metabolic imaging techniques like PET and FMRI work:

- they're *metabolic.* So, you're not looking at neural activity. You're looking at an index of oxygen consumption.

- they're *slow.* Metabolic processes unfold over seconds. Important neural events can occur on the order of milliseconds or less.

- they have only *fair* spatial resolution. FMRI is better than PET, but resolutions are on the order of millimeters, whereas important things are going on at micron scale.

Applying the logic of cognitive psychology to metabolic imaging data...



You can't just say “x part of the brain is active during y task,” because there's activity all over the brain all of the time.

So, there has to be some kind of control condition...

How did Posner *et al.* apply subtraction logic to isolate identification of abstract word and letter forms from semantic processing of word meaning?

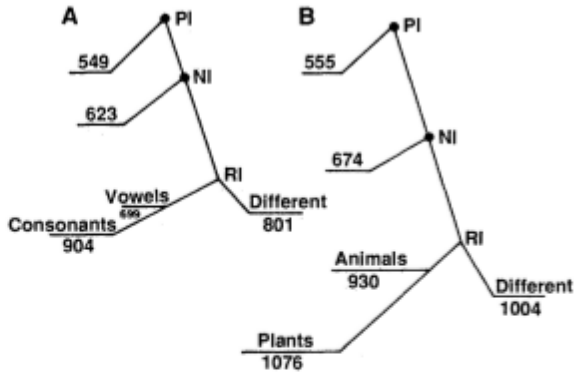


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Interestingly, the region they identify as involved in semantic processing of words corresponds roughly to Broca's area.

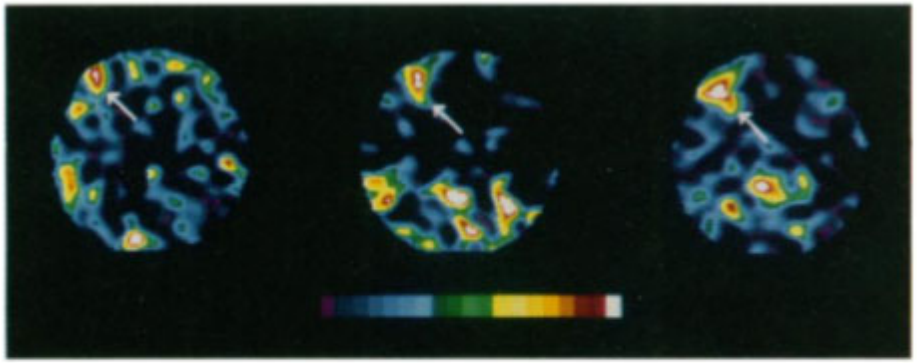
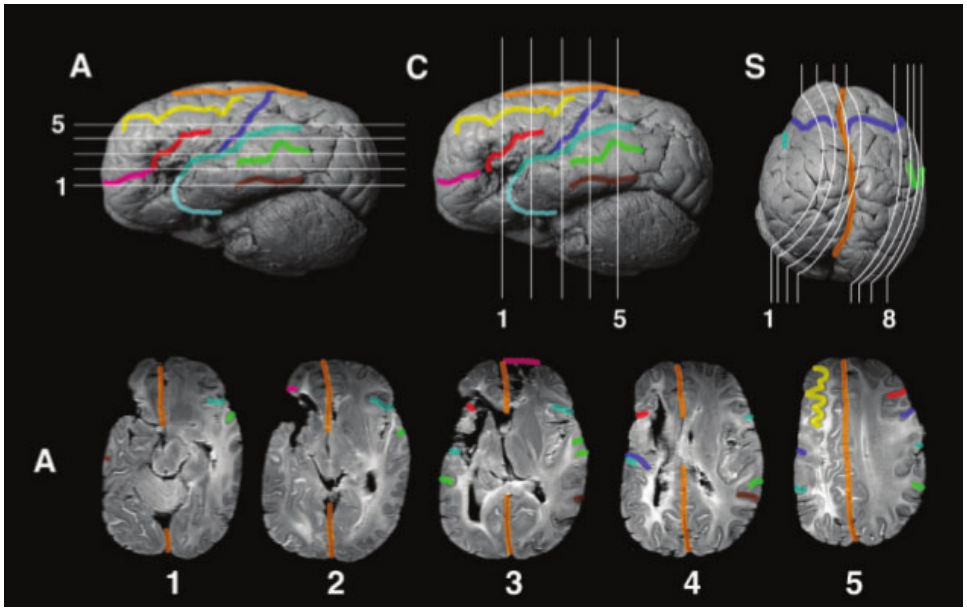


Fig. 4. Sample data from the PET activation studies. The arrows indicate areas of activation in the left inferior prefrontal cortex found active in all three semantic processing conditions. (**Left**) Monitoring visual words for dangerous animals (minus passive visual words). (**Middle**) Generating uses (minus repeat) for visual stimuli. (**Right**) Generating uses (minus repeat) for auditory stimuli. In each condition an area of cortical activation was found in the anterior cingulate gyrus on a higher slice (Fig. 3). The color scale indicates the relative strength of activation (purple indicates the minimum and white, the maximum, for that condition) (*26*).



A rhetorical question (but one worth exploring):
Is this better than phrenology?

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Fact or Phrenology?

The growing controversy over fMRI scans is forcing us to confront whether brain equals mind

By David Dobbs

A rhetorical question (but one worth exploring): Is this better than phrenology?

Conclusions

The PET data provide strong support for localization of operations performed on visual, phonological, and semantic codes. The ability to localize these operations in studies of average blood flow suggests considerable homogeneity in the neural systems involved, at least among the right-handed subjects with good reading skills who were used in our study.

The PET data on lexical access complement the lesion data cited here in showing that mental operations of the type that form the basis of cognitive analysis are localized in the human brain. This form of localization of function differs from the idea that cognitive tasks are performed by a particular brain area. Visual imagery, word reading, and even shifting visual attention from one location to another are not performed by any single brain area. Each of them involves a large number of component computations that must be orchestrated to perform the cognitive task.

Tying it all together...

Language has turned up at a number of key moments in debates about localization of function:

Broca's presentation of Tan helped establish the neuropsychological approach of lesion-behavior analysis.

Some of the very first PET and FMRI studies used activation of “Broca's Area” in language tasks as evidence that cognitive functions could be localized.

What about language makes it so well suited to these investigations?

Required for Tuesday:

Read Chomsky's review of Verbal Behavior http://www.chomsky.info/articles/1967----.htm

Optional, but really really interesting, reading (especially for Cog Sci majors): Peter Norvig's critique of Chomsky's thoughts on big data and AI: http://norvig.com/chomsky.html

This interview in The Atlantic, in which Chomsky lays out his position on AI (much of this is well, and quite fairly, summarized in Norvig's response): http://www.theatlantic.com/technology/archive/2012/11/noam-chomsky-on-where-artificial-intelligence-went-wrong/261637/

These Language Log posts about the argument between Chomsky and Norvig:http://languagelog.ldc.upenn.edu/nll/?p=3172 http://languagelog.ldc.upenn.edu/nll/?p=3180

REMINDER: SEND SPEECH ERRORS!!